

FINANCIAL INSTABILITY IN LATIN AMERICA: A MINSKIAN-KALECKIAN PERSPECTIVE

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Date received: February 15, 2012. Date accepted: June 12, 2012

Abstract

The same dynamics, namely the rise of a growth model oriented towards financial activities, have led to both the current financial crisis and global imbalances characterized by a rising public deficit in the United States. This work has two objectives. The first is to analyze how the flow of foreign investment and cross-border mergers and acquisitions affect the economy in the United States and Latin America. The second is to establish a theoretical framework of macroeconomic financial instability and apply it to examine the Latin American economy in the past decade. We propose a macroeconomic interpretation of the theory of financial instability, combined with a Minskian-Kaleckian theoretical framework.

Keywords: Cross-border mergers and acquisitions, income balance, financial speculation, rate of capital use, expected utility rate.

INTRODUCTION

The international economy is currently in a severe crisis, amidst confusion in the financial markets and the deterioration of productive activities. Favorable economic performance in Latin America, which has been driven by primary sector exports and the increase in foreign direct investment (FDI) following 2002, has also become a macroeconomic uncertainty. Most research attributes current difficulties to the economic situation of industrialized countries, that is, factors outside of the regional economy. This point of view explains that during the high growth phase, Latin American economic strategies did not lead to any problems. However, these excellent results have been destroyed by the contagion of the high-risk mortgage crisis in the United States.

This work reveals that financial instability was generated during the phase of high growth. The Minskian hypothesis of financial instability focuses on the vulnerability of non-financial entities in the face of a transforming financial situation. The goal of this work is to establish the theoretical framework of financial instability in Latin American macroeconomics, as payment of profits and interests abroad are increasing. In the first section, we identify the structure of the balance of payments. In the second part, we set up the financial instability model, based on the role of investment in neo-Kaleckian and post-Keynesian theory. The fall in the capacity utilization rate is one of the key factors determining investment and the dynamic adjustment model.

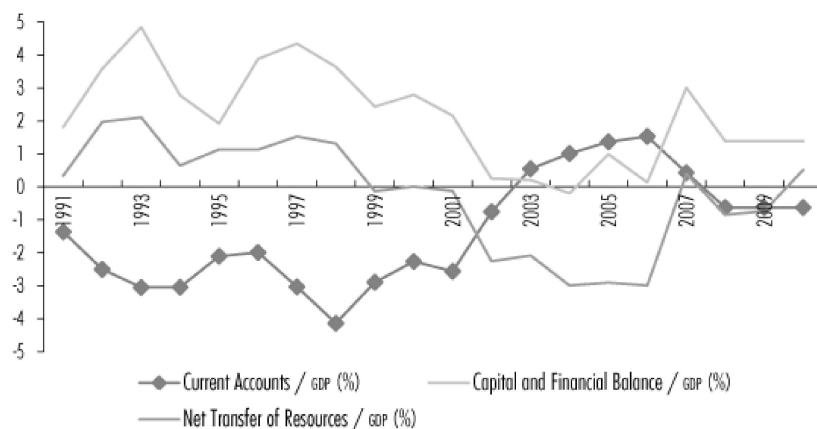
STRUCTURAL TRANSFORMATION IN THE BALANCE OF PAYMENT FOR LATIN AMERICA

The Latin American economy has undergone structural transformation throughout the last two decades. Before 2001, the balance of payment was composed of the public accounts deficit and the surplus of the capital and financial accounts, where inflows of FDI and foreign portfolio investment made up the surplus of this account.

Starting in 2002, the surplus of public accounts and the capital and financial balance stood out until 2007, while the net transfer of resources saw a negative trend (Figure 1). The region has faced consequences resulting from the rise in international prices for food and hydrocarbons. The increase in the terms of trade reached 28.1% in 2008 as compared to 2001, while the rate of variation fell to 2.6% in 2007 and 4.6% in 2008.¹ The net transfer of resources, shown in the *Preliminary Economic Balance of Latin America and the Caribbean*, published by the Economic Commission for Latin America and the Caribbean (ECLAC), is defined as: the net earnings of capital minus the net profits and interests demonstrated in the income balance. The negative figures for this indicator show that net payments of profits and interests were made abroad.

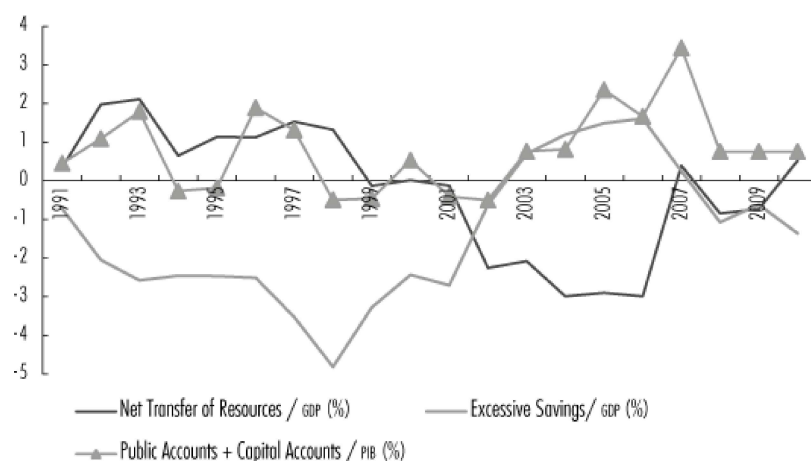
The surplus of current accounts reflects the fact that domestic savings exceed fixed investment (Figures 2 and 3). The generation of excessive liquidity due to the surplus of the balance of payments provoked an expansion in savings, which reached 23.3% of the Gross Domestic Product (GDP) in 2006. Authorities adopted a restrictive monetary policy due to the high interest rate, to reduce the risk of inflation. However, this strategy did not work as an anti-cyclical policy, because fixed investment also increased from 19% of GDP in 2003 to 23.8% in 2008.

Figure 1. Balance of Payments in Latin America, as a percentage of GDP (%)



Source: ECLAC. Preliminary Economic Balance of Latin America and the Caribbean 2011, p. 103; Preliminary Economic Balance of Latin America and the Caribbean 2005, p. 170; Economic Studies of Latin America and the Caribbean 2002-2003, p. 390.

Figure 2. Net Transfer of Resources and Excessive Savings as a Percentage of GDP (%)



Source: ECLAC. Preliminary Economic Balance of Latin America and the Caribbean 2011, p. 103; Preliminary Economic Balance of Latin America and the Caribbean 2005, p. 170; Economic Studies of Latin America and the Caribbean 2002-2003, p. 390.

Figure 3. Fixed Investment and Domestic Savings, as a Percentage of GDP (%)

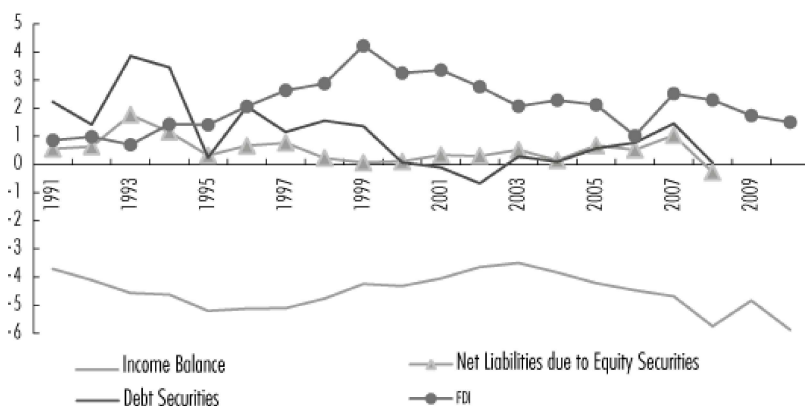


Source: ECLAC. Preliminary Economic Balance of Latin America and the Caribbean 2011, p. 103; Preliminary Economic Balance of Latin America and the Caribbean 2005, p. 170; Economic Studies of Latin America and the Caribbean 2002-2003, p. 390.

The rate of increase in international commerce fell from 7.1% in 2007 to 4.9% the following year. In this same time period, there was a net exit of financial resources, shown in the capital and financial balance, as a consequence of the reduced inflow of FDI and portfolio investment from industrialized economies.

Focusing on these external conditions, most hypotheses assume that: *i)* the economy of the region did not have any internal or external imbalance before the 2008 crisis, because this crisis originated in the external market; and *ii)* the differing levels of economic contraction among Latin American countries can be attributed to varying levels of commercial openness and export diversification.² In other words, according to this analysis, internal demand has registered a sustained level of growth before and after the crisis, as the GDP annual growth rate for the region recovered to 5.9% in 2009.

Figure 4. The Capital and Financial Balance of Latin America, as a percentage of GDP



Source: ECLAC. *Latin America and the Caribbean: Historical Series of Economic Data, Latin America, Balance of Payments 1980-2008; Balance of Payments in Dollars at Current Prices*, <http://www.ECLAC.cl/deype/cuaderno37/esp/index.htm> and Preliminary Economic Balance of Latin America and the Caribbean 2011, pp. 99, 110, 111.

The above allows us to identify a growth mechanism accompanied by the transfer of resources to outside of the region. Figure 3 indicates the evolution of FPI, the net liabilities per participation in capital (inflow of foreign portfolio investment and cross-border mergers and acquisitions), and the net liabilities per debt bonds. A certain structural change also stands out in the capital and financial balance, which allows us to conclude that cross-border mergers and acquisitions and debts have obligated the regional economy to make additional payments of profits and interests abroad.

Facing growth rates of 4.6% or 5.8% between 2003 and 2006, the monetary authorities of the regional countries applied an anti-cyclical policy by increasing the nominal interest rate. In reality, this had no effect on the level of fixed investment, although it did encourage private entities to issue debt bonds in the international market. A 2002 ECLAC document analyzes this issue, concluding that (following the global uncertainty of 2002) Latin American countries had to recover access to sources of foreign financing with normal interest rates. However, this implied restructuring external debt.³

This observation orients us towards examining the relationship between fixed investment, domestic savings and payments of profits and interests, in the years before the crisis. However, excessive savings and a surplus in the capital and financial balance cannot be explained by national accounts in the neoclassical theoretical framework. In the following section, we build the model of financial stability/instability, based on the Minsky hypothesis. THE MINSKIAN-KALECKIAN METHOD

Identifying the factors that determine fixed investment is essential to understanding economic growth. Most research on Latin American economics analyzes how the low rate of domestic savings has obligated the region to depend on external savings to finance investment, which is why foreign vulnerability has been an obstacle to growth.⁴

In the Kaleckian model, investment and savings are explained in different contexts. The role of investment includes the explanatory variables of the profit rate and the capacity utilization rate, and the savings rate is determined by the propensity of businessmen and income distribution. This school of thought says that a low propensity to save increases the rate of investment and capital accumulation. Post-Keynesian theory teaches that investment produces a flow of additional savings that finances itself, and as such, the *ex ante* level of savings does not restrict the investment rate. In this theoretical framework, the main cause of growth vulnerability is the oligopolized regime of the financial sector and the existence of related credits.

Investment and the Capacity Utilization Rate in the Kaleckian Model

The Kaleckian model is built on three functions: the function of fixed investment I , the function of savings S , and the capacity utilization rate u , defined as: $u = Y/K$, where Y represents national income and K is the formation of fixed capital.⁵ Growth in the short and medium term is based on the additional utilization of resources and capacities that have been used at a lower level than their maximum utilization. In general, the function of fixed investment is illustrated in the following manner, with the parameters of: $g_1 - g_4$.⁶



where r_{sf} corresponds to the expected profit rate for investors (shareholders), u is the capacity utilization rate, W/K registers Tobin's Q , where W is the level of capitalization in the market, and D is external debt. In the securitization phase, the effect of Tobin's Q on investment is not identified. If businessmen intervene in the stock market and buy and sell stocks from their own entities for the business fund, the rise in Tobin's W does not result in additional fixed investment. So, ignoring Tobin's W , we present the accumulation function in the following manner:

$$g = I/K = g_0 + g_1(u) + g_2 p^* \quad (1)$$

In (1), p^* is the expected profit rate. The traditional Kaleckian model assumes an adjustment to the capacity utilization rate in the medium and long term, until reaching the "normal" rate $\bar{u}$. Additional investment produces increased income, which is divided into salary income for employees and business profits. Given the assumption that salary income does not generate savings, income distribution formulates the savings function. It depends on the adjustment parameter of u and the income distribution, and guarantees a stable or unstable regime in the balance between investment and savings.

The purpose of the neo-Kaleckian theory is to illustrate the capacity utilization rate as the endogenous variable in the model. Lavoie (1992) demonstrates the capacity utilization rate function built by the explanatory variables of real salary and business earnings:⁷

$$u = \frac{g\left(\frac{v}{s_p}\right) + f\left(\frac{W/p}{y_v}\right)}{1 - \left(\frac{W/p}{y_v}\right)}$$

where:

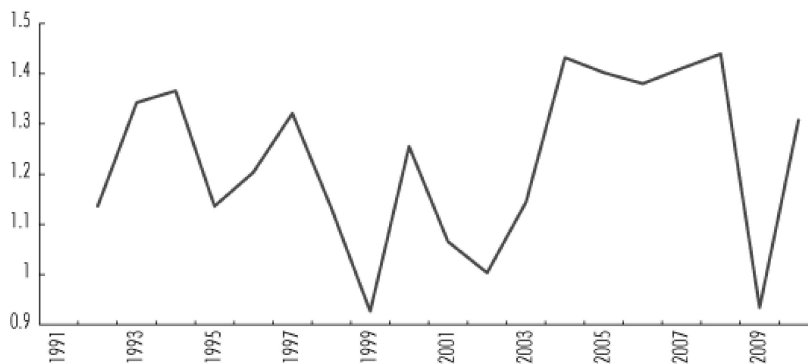
W/p represents the real salary rate

y_v is labor productivity, and

$v = \frac{\text{capital}}{\text{maximum capacity}}$ is the propensity to save

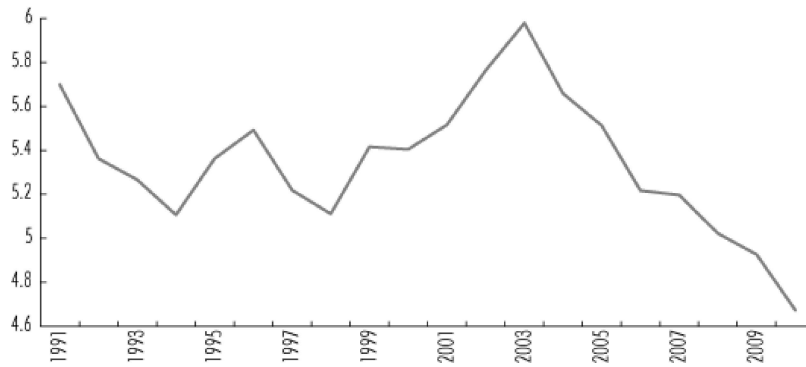
Oreiro (2004) analyzes the fact that oligopolistic entities tend to maintain excessive capacity to keep their level of production in the face of others. Lima and Meirelles (2007) show that an adjustment in capacity utilization affects the *mark-up* level of the interest rates of banks. If the increase in the capacity utilization rate reflects the possibility to generate additional profits, the banks prefer to lower the interest rate to support additional investment. On the other hand, if companies do not have the expected rate of high profits, the increased utilization rate causes the bank risk, as they will require a higher interest rate.

Figure 5. Rate of Accumulation [$I/K_{(-1)}$]



Source: ECLAC, *Latin America and the Caribbean: Historical Series of Economic Data > 5; Latin America and the Caribbean, National Accounts in Dollars at Current Prices, 1990-2008 > 5.2; Gross Domestic Product by Type of Expense*, <http://www.eclac.cl/deype/cuaderno37/datos/5.2.1.xls>; Preliminary Economic Balance of Latin America and the Caribbean 2011, p. 102.

Figure 6. Capacity Utilization Rate (%)



Source: ECLAC, *Latin America and the Caribbean: Historical Series of Economic Data > 5; Latin America and the Caribbean, National Accounts in Dollars at Current Prices, 1990-2008 > 5.2; Gross Domestic Product by Type of Expense*, <http://www.eclac.cl/deype/cuaderno37/datos/5.2.1.xls>; Preliminary Economic Balance of Latin America and the Caribbean 2011, p. 102.

Based on this research and the observations in Figures 5 and 6, we can determine the function of capital accumulation and domestic savings. Let us consider an economy in the phase of securitization (financialization), where business profits are divided between company income whose propensity for saving is s_e , and domestic investors (*shareholders*) whose propensity for saving is s_c . rD represents the payments of profits and interests abroad for debts and investments, as rD corresponds to the net transfer of resources abroad.⁸ Assuming that domestic businessmen channel their net profits ($P - rD$) into savings and/or dividends for investors, total domestic savings are as follows:⁹

$$S = s_e (P - rD) + s_c (1 - s_e) (P - rD) \quad (2)$$

Where P is the profit generated by the investment. Let us assume that companies plan their "planned" level of investment I^* :

$$I^* = \beta s_e (P - rD) + I_0, \text{ where } \beta (0 < \beta < 1) \text{ is the investment parameter} \quad (3)$$

Assuming that planned investment corresponds to the balance between aggregate supply and aggregate demand, we solve $I^* = S$ and obtain:

$$\pi = \frac{1}{a_1 (1-f)} \left(\frac{I_0}{K} \right), \quad (4)$$

where:

$$a_1 = s_c (1 - s_e) + s_e (1 - \beta) > 0,$$

$$f = rD/P, \quad \pi = P/K$$

The endogenous adjustment to the level of accumulation is illustrated as a function of the difference of the planned accumulation rate and the current rate, and as such, the dynamic adjustment is:

$$\dot{g} = \alpha (g^* - g), \text{ where } \alpha (>0) \text{ is the adjustment parameter} \quad (5)$$

By definition,

$$g^* = \beta s_e (P/K - rD/K) + g_0 = \beta s_e (1-f) \pi + g_0 \quad (6)$$

So, substituting (6) in (5), we obtain the following:

$$\dot{g} = \alpha (g^* - g) = \alpha \left[\left(\frac{\beta s_e}{a_1} \right) + g_0 - g \right] \quad (7)$$

The capacity utilization rate is considered the endogenous variable in the previous section, as we do not propose a balance between investment and savings by adjusting this rate. Moreover, we identify the function of savings and that of payment of profits and interests abroad, based on the Minskian financial instability hypothesis. Minsky's theory focuses on the excess and/or scarcity of the cash flow generated by investment, in comparison with the balance of interest payments

to the external fund to finance it. The function of investment in the neoclassical theoretical framework is based on the hypothesis that the behavior of sources of investment financing does not affect the level of investment, and as such, financing from external funds does not have a negative influence on the level of additional investment. The key point of the Minskian theory is that the risk for the debtor and the lender is elevated, in the same way that investment financing depends on an external fund, as well as the issuance of bonds and the raising of bank loans.

At the beginning of a boom in the economic cycle, companies tend to finance investments with their own resources, and this is called *hedge* finance. Additional investment brings about the need to raise external funds. High interest rate debts lead to the need for companies to obtain some fund with high liquidity before being able to capture profit flows. This obliges companies to refinance their debts for new external funds and the finance in this situation is classified as *speculative* finance. If most companies are at the stage of speculative finance, some will need to sell their assets to have liquidity. This phase is defined as *Ponzi* finance.¹⁰

Minsky's analysis focuses on the balance of a representative entity, and as such, we must adopt a different point of view to apply this to macroeconomic research. If an entity is in the stage of speculative or *Ponzi* finance, another may transfer its funds to the former in order to not increase the risk of the debtor. In emerging economies, companies tend to depend on the issuance of bonds and raising bank loans in the international market to start meeting the need for additional investment. On the other hand, small entities do not have access to the international financial market and are also excluded from the domestic market. As such, they must continue their financing with their own resources. The increase in net liabilities due to the issuance of debt bonds and capital participation bonds brings about the need to make payments of profits and interests. This obliges some entities to take on even more debt. Even when investment is increasing and the economy has a high growth rate, the need to pay profits and interests abroad introduces the risk of uncertainty for the real exchange rate.

The risk of the debtor in the macroeconomic context is interpreted in this way. The hypothesis of this work is that financial instability in emerging economies is defined as a situation where additional investment does not produce accumulated profits in the domestic economy, as a consequence of increased payments of profits and interests abroad. In these circumstances, additional investment brings about the need to take on additional debt to pay interest, as it increases the risk of devaluation and/or capital flight. Our analysis is supported by Foley (2003), which presents the function of investment with the explanatory variable of the interest rate, and explains external debt as an increased function of the domestic interest rate. A high interest rate leads to elevated investment costs and higher payments for debt at the same time.

Profit P + additional debt $D \equiv$
Additional investment I + payment of interest on debt V

$$g = I/K, i = V/B, \pi = P/K$$

On the macroeconomic level, the company at stage $V + I > R \geq V$, where $D \geq 0$ and $D < 1$, is in the speculative stage of financing. A company with $R < V$, and as such $D > 1$, is at the *Ponzi* financing stage.

In macroeconomics, debt corresponds to the surplus of the capital and financial balance (= deficit of current accounts) and $D/K = I/K - sp(Y/K)$, where s represents the propensity to save of businessmen, with the assumption that consumption and savings of the businessmen come from profits P and $p = P/Y$. In the space (g, i) that demonstrates the different financial conditions, the area of $\pi > g > i$ and $\pi > i > g$ correspond to the hedge financing stage, $g > \pi > i$ indicate speculative financing and $g > i > \pi$, $i > g > \pi$ and $i > \pi > g$ demonstrate the economics of Ponzi financing. In the Foley model, the function of the surplus of the financial capital account D and investment I are functions of the interest rate, and as such, capital accumulation and payment of interest abroad are affected at the same time by the interest rate.

Because the average level of the interest rate for countries in the region is not given, we can take a look at the approximate evolution of financing in Figure 7, which indicates the annual rate of variation for: capital accumulation (I/K) domestic savings (S/K) and the net liabilities of the income balance. The data for the income balance are negative during this time period, as their rate of negative variation indicates the increase in payment of profits and interests abroad. With the assumption that savings reflect a part of the profit P , we can compare the rate of variation of I/K and savings. The rate of increase in savings drastically falls between 2004 and 2005, as we observe that $g \geq \pi$ between 2004 and 2009.

Figure 7. Annual Rate of Variation of g , s and the Income Balance Deficit (%)



Source: Prepared by the author based on ECLAC, Latin America and the Caribbean: Historical Series of Economic Data > 5; Latin America and the Caribbean, National Accounts in Dollars at Current Prices, 1990-2008 > 5.2; Gross Domestic Product by Type of Expense, <http://www.ECLAC.cl/deype/cuaderno37/datos/5.2.1.xls>; Preliminary Economic Balance of Latin America and the Caribbean 2011, pp. 99 and 102.

Foley's model (2003) cannot be applied to Latin American research from the last decade for a few reasons. First, the increase in the payment of profits and interests abroad, which can be identified by the deficit of the net transfer of resources, corresponds to a surplus in current accounts and capital and financial accounts. It is assumed that the inflow of FDI and portfolio investment have stimulated, at least in part, capital accumulation, and on the other hand, have supported the increased payment of profits and interests abroad. Second, since the 1990s, restrictive monetary policy has not had a negative effect on the evolution of fixed investment in the region. To demonstrate this idea in our model, we present the function of accumulation in equation (1) with the determining factor of the capacity utilization rate.

Moreover, this work addresses the effect of securitization on the economy. We consider that profits are divided into domestic savings for companies, domestic savings of investors and the payment of profits and interests abroad. With this assumption, we showed the savings function:

$$S = s_e (P - rD) + s_c (1 - s_e) (P - rD) \quad (2)$$

For a more detailed analysis, we must mention that the payment of profits and interests is not determined only by the interest rate, because the payment of profits that come from FDI and foreign portfolio investment are determined as a consequence of a business and investment strategy. Still, with the goal of simplifying the model, we use (2) in this section and we will return to this issue later on. Modifying the definition of f from (4), we obtain the function differentiated by time:

$$f = \frac{rD}{Y} \times \frac{Y}{P} = tu/\pi, \text{ where:}$$

$$t = rD/Y, \quad (7)$$

$$u = Y/K,$$

$$\pi = P/K = \frac{1}{a_1(1-f)}$$

$$\dot{f} = \dot{t} + \dot{u} - \dot{\pi} = \gamma \dot{t}(\dots) + \dot{u}(\dots) - \varepsilon \frac{1}{a_1(f^2 - 2f + 1)} = \gamma \dot{t}(\dots) + \dot{u}(\dots) - \varepsilon F(f), \quad (9)$$

Where γ and ε are the parameters that demonstrate the speed of adjustment.

Capital Accumulation and Payments Abroad

Using the function differentiated by time and shown in the previous sections, we present the dynamic adjustment model for the system.

$$\dot{g} = \alpha(g^* - g) = \alpha \left[\left(\frac{\beta s_e}{a_1} \right) + g^* - g \right], \text{ where:} \quad (7)$$

$$a_1 = s_c(1 - s_e) + s_e(1 - \beta) > 0$$

$$\dot{f} = \dot{t} + \dot{u} - \dot{\pi} = \gamma t(\dots) + \dot{u}(\dots) - \varepsilon F(f) \quad (9)$$

Linearizing equation 7 and 9 around the initial balance of g and f , we obtain the following matrix:

$$\begin{bmatrix} \dot{g} \\ \dot{f} \end{bmatrix} = \begin{bmatrix} \partial \dot{g} / \partial g & \partial \dot{g} / \partial f \\ \partial \dot{f} / \partial g & \partial \dot{f} / \partial f \end{bmatrix} \times \begin{bmatrix} g' \\ f' \end{bmatrix}, \quad (10)$$

Whose Jacobian matrix is made up of the following elements:

$$\partial \dot{g} / \partial g = -\alpha < 0 \text{ and assuming that the propensity } S_e \text{ is not affected by } f, \partial \dot{g} / \partial f = 0$$

With the assumption that t and u are not a function of g , we obtain:

$$\partial \dot{f} / \partial g = 0 \text{ and } \partial \dot{f} / \partial f > 0 \text{ when } f > 1 \text{ o } f \approx 1 \text{ and } \partial \dot{f} / \partial f < 0 \text{ when } f \ll 1.$$

The condition required for the convergent dynamic adjustment that indicates stability of the equilibrium is obtained by:

$$\begin{aligned} \text{Tr. } \partial \dot{g} / \partial g + \partial \dot{f} / \partial f &< 0 \text{ when } f \ll 1, \text{ and} \\ \partial \dot{g} / \partial g + \partial \dot{f} / \partial f &> 0 \text{ when } f \geq 1; \end{aligned} \quad (11)$$

$$\begin{aligned} \text{Det. } \left(\partial \dot{g} / \partial g \right) \left(\partial \dot{f} / \partial f \right) - \left(\partial \dot{g} / \partial f \right) \left(\partial \dot{f} / \partial g \right) &> 0 \text{ when } f \ll 1, \text{ and} \\ \left(\partial \dot{g} / \partial g \right) \left(\partial \dot{f} / \partial f \right) - \left(\partial \dot{g} / \partial f \right) \left(\partial \dot{f} / \partial g \right) &< 0 \text{ when } f \geq 1. \end{aligned} \quad (12)$$

We consider the result when $\partial \dot{g} / \partial f \neq 0$, $\partial \dot{t} / \partial g \neq 0$ and $\partial \dot{u} / \partial g \neq 0$.

For equations (4) and (7), the result is: $\partial \dot{g} / \partial f < 0$ when $\partial \dot{s}_e / \partial f > 0$ and/or $\partial \dot{s}_c / \partial f > 0$

These correspond to the case where, given an increase in the payment of profits and interest abroad, both domestic investors and businessmen have an increased propensity to save. In emerging economies, it is natural for the rate of increase of accumulation to exceed the rate of increase for domestic savings, as we consider the following: $\partial \dot{t} / \partial g > 0$.

In an economy where there is productive capacity that is not being used, the following is generated: $\partial \dot{u} / \partial g > 0$.

However, if we consider that capacity utilization is the endogenous variable as in Orebro (*Op cit.*) it is possible that oligopolistic companies prefer to advance the accumulation of capital by reducing their level of capacity utilization and increasing the mark-up rate on prices. For (12), $\partial \dot{g} / \partial f < 0$, $\partial \dot{t} / \partial g > 0$, and $\partial \dot{u} / \partial g < 0$ lead to the higher possibility that $\text{Det.} > 0$ when $f \geq 1$.

The total profit data for Latin America is not made explicit. Figures 1 and 2 indicate the expansion of payments of profits and interest abroad starting in 2002, as well as the evolution of fixed investment and domestic savings. Figure 7 shows the high probability for the increase in profits. Comparing the evolution of these factors allows us to assume that starting in 2002, $f \approx 1$ was established, generating financial speculation. Moreover, the following conditions were of note:

- Figure 5 indicates the increased evolution of the rate of accumulation until 2008, while the capacity utilization rate saw decreasing change starting in 2003.
- As a result of financialization, the flow of transferring profits and utilities abroad has grown. Decreased revenue is a result of the higher propensity for businessmen and domestic investors to save.

In this way there is a chance that between 2002 and 2008, the necessary conditions for an outgoing adjustment dynamic were created. In the high growth phase led by exports and FDI, the Latin American economy was in a stage of financial speculation and/or Ponzi financing, which brought about macroeconomic risk due to adjusting the capacity utilization rate.

The *World Investment Report 2011*, published by the United Nations Conference on Trade and Development (UNCTAD) and Safarian (2011) reveals that in the global economy nearly 50% of FDI was in the form of cross-border mergers and acquisitions from institutional investors between 2003 and 2008 leading to an increase in the stock market index. This produces a situation whereby additional profits must be paid to foreign investors, also during a time of decreased interest rates in the global market.

FINAL CONCLUSIONS

The composition of the balance of payments in Latin America has been in a unique position since 2002, and the payment of profits and interests abroad has been a defining element. The favorable evolution of fixed investment also brought about excessive domestic savings, while the capacity utilization rate continued to decrease. We have adopted the Kaleckian investment function and Minsky's theory of financial instability. Macroeconomic financial instability in emerging markets reflects an increase in risk for the debtor as a result of the growth of payments of profits and interest abroad. This research allows us to discern the high probability of an outgoing adjustment dynamic, which corresponds to instability in the balance generated by Ponzi financing.

BIBLIOGRAPHY

- Charles, Sébastien, "A Post-Keynesian Model of Accumulation with a Minskyan Financial Structure", *Review of Political Economy*, vol. 20, no. 3, 2008, pp. 319-331.
- Comisión Económica para América Latina y El Caribe, Balance preliminar de la economía de América Latina y El Caribe 2005, Santiago de Chile, CEPAL, various numbers.
- Foley, Duncan K., "Financial Fragility in Developing Economies", in Dutt, Amitava Krishna and Jaime Ros (eds.), *Development Economics and Structuralist Macroeconomics Essays in Honor of Lance Taylor*, New York, Edward Elgar, 2003.
- Hein, Eckhard and Till van Treeck, "'Financialization' in Post-Keynesian Models of Distribution and Growth: a Systematic Review", in Setterfield, M. (ed.), *Handbook of Alternative Theories of Economic Growth*, Cheltenham, Edward Elgar, 2010.
- Hein, Eckhard, Marc Lavoie and Till van Treeck, "Some Instability Puzzles in Kaleckian Models of Growth and Distribution: a Critical Survey", *Cambridge Journal of Economics*, vol. 35, no. 3, Oxford, 2011, pp. 587-612.
- Lavoie, Marc, *Foundations of Post-Keynesian Economic Analysis*, 1992, Hants/Vermont, Edward Elgar.
- Lima, Gilberto Tadeu and Antonio V. A. Meirelles, "Macrodynamics of Debt Regimes, Financial Instability and Growth", *Cambridge Journal of Economics*, vol. 31, Oxford, 2007, pp. 563-580.
- Minsky, Hyman P., *Stabilizing an Unstable Economy*, New Haven and London, Yale University Press, 1986.
- Oreiro, José Luis, "Accumulation Regimes, Endogenous Resired Rate of Capacity Utilization and Income Distribution", *Investigación Económica*, vol. LXIII, no. 248, Mexico, FCE-UNAM, 2004, pp. 41-70.
- Safarian, A. Edward, "International Mergers and Acquisitions", in Jvanovic, Miroslav N. (ed.), *International Handbook on the Economics of Integration, Volume III, Factor Mobility, Agriculture, Environment and Quantitative Studies*, Cheltenham/Northampton, Edward Elgar, 2011.
- United Nations Conference on Trade and Development, *World Investment Report 2009*, New York, UNCTAD, 2011.

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¹ ECLAC, *Preliminary Economic Balance of Latin America and the Caribbean 2008*, p. 70.

² ECLAC, *Preliminary Economic Balance of Latin America and the Caribbean 2009*, p. 59.

³ ECLAC, *Preliminary Economic Balance of Latin America and the Caribbean 2002*, p. 16.

⁴ ECLAC, *An Economic Study of Latin America and the Caribbean 2007*, p. 41.

⁵ Lavoie, 1992, p. 301 and Hein, Lavoie and Treck, 2011.

⁶ Hein and Treck, 2010.

⁷ Lavoie, 1992, p. 301.

⁸ Here *D* includes additional external debt and profits produced by FDI, among other items.

⁹ This model is supported by Charles, 2008.

¹⁰ Minsky, 1986, pp. 190-196.